Sample Data Insertion

Realistic seed rows for all nine tables in the library database

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Project: Library Management System. **Output convention:** result tables below are reproducible expected outputs from the fixed rows in the shared SQL script. They are not screenshots of MySQL execution.

1. Sample-data plan

The dataset is deliberately small enough to inspect but varied enough to support joins and aggregate queries. It includes books with multiple authors, two copies of one book, returned and open loans, a member with no loans, one unborrowed title, and reservations in three states.

Table	Rows	Purpose in the reports
publishers	3	Compare publisher references across books.
authors	8	Represent eight author identities.
categories	4	Support grouped catalog summaries.
books	8	Provide different categories and prices.
book_authors	9	One book has two authors.
book_copies	10	Track physical copies, including two copies of book 1 and book 8.
members	5	Include varied circulation histories and one non-borrower.
loans	8	Include 5 open loans and 3 returned loans.
reservations	4	Include pending, fulfilled, and cancelled requests.

2. Insert statements

The following statements populate every table and are shared verbatim with the unified SQL script. The assignment dates are fixed; overdue examples in Assignments 8 and 9 use the fixed comparison date 2026-10-23.

```
INSERT INTO publishers (publisher_id, publisher_name, city) VALUES
    (1, 'Northstar Press', 'Bengaluru'),
    (2, 'Meridian Academic', 'Mumbai'),
    (3, 'Cedar House', 'Kolkata');

INSERT INTO authors (author_id, author_name, country) VALUES
    (1, 'Ada Sen', 'India'),
    (2, 'Ravi Menon', 'India'),
    (3, 'Mira Das', 'India'),
    (4, 'Arjun Rao', 'India'),
    (5, 'Lila Fernandes', 'India'),
    (6, 'Omar Khan', 'India'),
    (7, 'Neha Iyer', 'India'),
    (8, 'Dev Mukherjee', 'India');

INSERT INTO categories (category_id, category_name) VALUES
    (1, 'Database'),
    (2, 'Programming'),
    (3, 'Fiction'),
    (4, 'History');

INSERT INTO books
    (book_id, isbn, title, publication_year, publisher_id, category_id, list_price)
VALUES
    (1, '9780133970777', 'Relational Database Basics', 2022, 2, 1, 650.00),
    (2, '9789355320216', 'SQL for Beginners', 2023, 1, 1, 520.00),
    (3, '9781492051367', 'Python in Practice', 2024, 2, 2, 720.00),
    (4, '9788194907132', 'The River Beyond', 2021, 3, 3, 480.00),
    (5, '9788129145536', 'Data Structures Made Clear', 2020, 1, 2, 590.00),
    (6, '9789353040131', 'A Short History of Bengal', 2019, 3, 4, 430.00),
    (7, '9780596004841', 'Learning Web Design', 2022, 1, 2, 680.00),
```

```
    (8, '9789387779261', 'Mystery at Lake Road', 2025, 3, 3, 550.00);

INSERT INTO book_authors (book_id, author_id, author_role) VALUES
    (1, 1, 'Author'),
    (1, 2, 'Author'),
    (2, 1, 'Author'),
    (3, 3, 'Author'),
    (4, 4, 'Author'),
    (5, 5, 'Author'),
    (6, 6, 'Author'),
    (7, 7, 'Author'),
    (8, 8, 'Author');

INSERT INTO book_copies
    (copy_id, book_id, accession_code, shelf_location, copy_status)
VALUES
    (1, 1, 'ACC-001', 'DB-A1', 'ON_LOAN'),
    (2, 1, 'ACC-002', 'DB-A1', 'ON_LOAN'),
    (3, 2, 'ACC-003', 'DB-A2', 'AVAILABLE'),
    (4, 3, 'ACC-004', 'PR-B1', 'ON_LOAN'),
    (5, 4, 'ACC-005', 'FI-C1', 'AVAILABLE'),
    (6, 5, 'ACC-006', 'PR-B2', 'ON_LOAN'),
    (7, 6, 'ACC-007', 'HI-D1', 'AVAILABLE'),
    (8, 7, 'ACC-008', 'PR-B1', 'ON_LOAN'),
    (9, 8, 'ACC-009', 'FI-C2', 'AVAILABLE'),
    (10, 8, 'ACC-010', 'FI-C2', 'AVAILABLE');

INSERT INTO members (member_id, member_no, full_name, email, phone, joined_on) VALUES
    (1, 'LIB001', 'Asha Mehta', 'asha.mehta@example.edu', '+91-90000-00001', '2025-06-10'),
    (2, 'LIB002', 'Kabir Shah', 'kabir.shah@example.edu', '+91-90000-00002', '2025-07-02'),
    (3, 'LIB003', 'Sara Nair', 'sara.nair@example.edu', '+91-90000-00003', '2025-08-18'),
    (4, 'LIB004', 'Dev Bose', 'dev.bose@example.edu', '+91-90000-00004', '2025-09-01'),
    (5, 'LIB005', 'Isha Rao', 'isha.rao@example.edu', '+91-90000-00005', '2025-09-15');

INSERT INTO loans (loan_id, copy_id, member_id, loan_date, due_date, return_date) VALUES
    (1001, 1, 1, '2026-10-01', '2026-10-15', NULL),
    (1002, 2, 2, '2026-10-10', '2026-10-24', NULL),
    (1003, 3, 3, '2026-10-02', '2026-10-16', '2026-10-15'),
    (1004, 4, 1, '2026-10-05', '2026-10-19', NULL),
    (1005, 5, 4, '2026-09-20', '2026-10-04', '2026-10-03'),
    (1006, 6, 3, '2026-10-08', '2026-10-22', NULL),
    (1007, 7, 2, '2026-09-30', '2026-10-14', '2026-10-14'),
    (1008, 8, 3, '2026-10-12', '2026-10-26', NULL);

INSERT INTO reservations
    (reservation_id, member_id, book_id, reservation_date, reservation_status)
VALUES
    (2001, 5, 8, '2026-10-19', 'PENDING'),
    (2002, 3, 1, '2026-10-02', 'FULFILLED'),
    (2003, 2, 5, '2026-10-03', 'CANCELLED'),
    (2004, 1, 8, '2026-10-20', 'PENDING');
```

3. Verify table counts

Run this after all inserts. The expected counts confirm that every table received the intended rows.

```
SELECT 'publishers' AS table_name, COUNT(*) AS row_count FROM publishers
UNION ALL SELECT 'authors', COUNT(*) FROM authors
UNION ALL SELECT 'categories', COUNT(*) FROM categories
UNION ALL SELECT 'books', COUNT(*) FROM books
UNION ALL SELECT 'book_authors', COUNT(*) FROM book_authors
UNION ALL SELECT 'book_copies', COUNT(*) FROM book_copies
UNION ALL SELECT 'members', COUNT(*) FROM members
UNION ALL SELECT 'loans', COUNT(*) FROM loans
UNION ALL SELECT 'reservations', COUNT(*) FROM reservations;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

table_name	row_count
publishers	3
authors	8
categories	4
books	8
book_authors	9
book_copies	10
members	5
loans	8
reservations	4

4. Quick copy-status check

The copy status count confirms that five copies are currently checked out and five remain available.

```
SELECT copy_status, COUNT(*) AS copy_count
FROM book_copies
GROUP BY copy_status
ORDER BY copy_status;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

copy_status	copy_count
AVAILABLE	5
ON_LOAN	5

5. Screenshot steps

In MySQL Workbench, run the insert block after Assignment 5 has created the schema. Select all rows from a small table such as `members` and capture the SQL editor and Result Grid. For a compact overview, capture the row-count query above. The report tables are expected results, not execution screenshots.